

Phase 3: Articulation

The pedagogical core: do the three statements actually tie together?

The four checks, and the accounting idea behind each

Check	What it verifies
Balance sheet balances	Assets = Liabilities + Equity, always, by definition
Retained earnings roll-forward	$RE_{end} = RE_{begin} + \text{Net income} - \text{Dividends}$
Cash ties out	Ending cash on the cash flow statement = cash on the balance sheet
Indirect-method reconciliation	$\text{Net income} + \text{non-cash items} \pm \text{working capital} = \text{cash from ops}$

Every check reports a signed dollar delta scaled against a 1%-of-base tolerance -- never a pass/fail boolean. That's deliberate: a check that fails on real data is showing you something true about the filing or about a gap in `normalize.py`'s tag map, and hiding it behind a green checkmark would throw away the lesson.

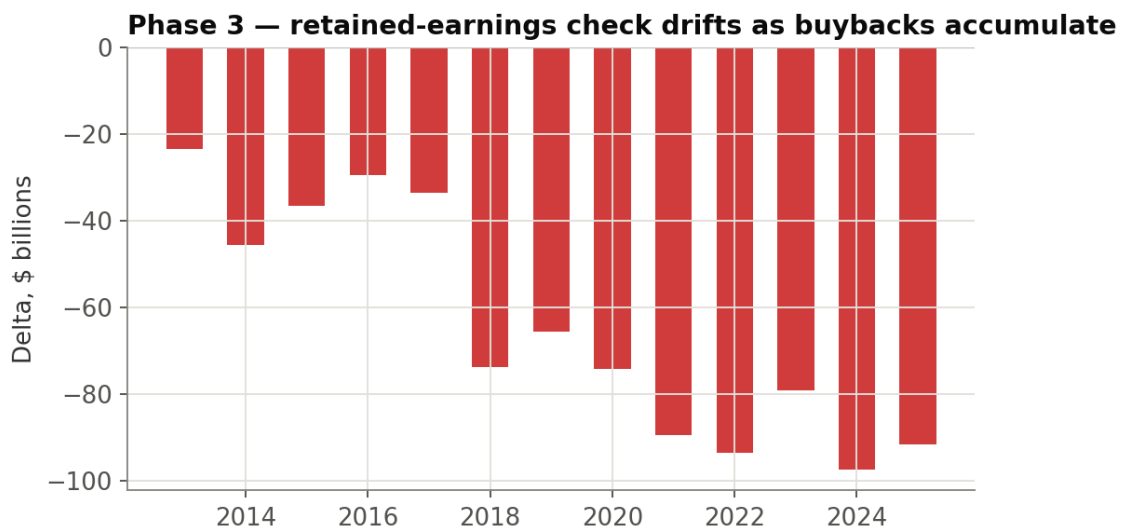
The accruals lesson, specifically

The fourth check is the heart of the exercise. Net income is an accrual number: it recognizes revenue when earned and expenses when incurred, not when cash actually moves. The indirect method reconciles the two by adding back non-cash charges (depreciation, stock compensation) and adjusting for the timing gaps in working capital -- if receivables went up, some of this period's revenue hasn't been collected in cash yet, so that increase is subtracted; if payables went up, some expenses haven't been paid in cash yet, so that increase is added back. This tool recomputes those deltas straight from two balance sheets, rather than trusting a company's own tagged "change in receivables" figure, specifically so the two can be checked against each other.

A real, large delta that is a lesson, not a bug

Apple's retained-earnings check shows a delta of -\$91.7 billion in FY2025, and the gap has grown every year since Apple started paying dividends in 2012. This is not a tagging error: Apple retires the shares it repurchases and charges the excess over par value against retained earnings, rather than holding them in a separate treasury-stock account. The textbook formula $RE_{end} = RE_{begin} + NI - \text{Dividends}$ has no term for that, so the delta grows every year buybacks happen -- and the size of the delta is itself a rough measure of how much share retirement is charged against retained earnings that period.

STATEMENT ANALYZER — PHASE 3



Apple's retained-earnings roll-forward delta, FY2013-FY2025 (no delta is computable before FY2013 -- Apple paid no dividends before August 2012, so "dividends paid" is genuinely unmapped, not zero).

KEY TAKEAWAY

A large articulation delta on real data is usually teaching you something real -- either about the filer's specific accounting choices, or about a line item your tag map hasn't reached yet. The check's job is to surface that gap clearly, with a real dollar number attached, not to decide whether it's "okay."